

Universal Excel Data Cleaner

VBA Automation • Setup • One-Click Use • Interview Demo

1. What we built

Universal Excel Data Cleaning & Quality Automation using VBA. The macro lives in **PERSONAL.XLSB** and runs on the active workbook. A Quick Access Toolbar button gives a one-click workflow.

2. What it does

- Cleans extra spaces and common text inconsistencies.
- Detects missing values and exact duplicate records.
- Checks email and phone fields when those columns exist.
- Detects and formats date-related fields.
- Cleans common financial fields such as salary, revenue, amount, price and cost.
- Adds Data_Quality_Flag values instead of inventing missing business data.
- Creates Clean_Data and Data_Quality_Report in the active workbook.

3. One-time setup

1. Create PERSONAL.XLSB: View → Macros → Record Macro → Store macro in: Personal Macro Workbook → OK → Stop Recording.
2. Open VBA: Developer → Visual Basic → Project Explorer → VBAProject (PERSONAL.XLSB).
3. Insert a Module and paste the complete code below.
4. Debug → Compile VBAProject. If successful, Ctrl+S.
5. Close Excel and save PERSONAL.XLSB when prompted.

4. Add the one-click button

- Quick Access Toolbar ▼ → More Commands.
- Choose commands from: Macros.
- Select PERSONAL.XLSB!Universal_Data_Cleaner → Add.
- Modify the icon/name to **Clean Data** → OK.

5. Use on any new Excel file

1. Open the new dataset.
2. Activate the sheet containing the data.
3. Click **Clean Data** on the Quick Access Toolbar.
4. The macro creates **Clean_Data** and **Data_Quality_Report** in that workbook.
5. Review flagged records before using the cleaned output.

6. Interview demo

Demo: Open a messy dataset → click Clean Data → show Clean_Data → show Data_Quality_Report → repeat on a different dataset.

Interview line: “I built a reusable Excel VBA data-cleaning automation stored in PERSONAL.XLSB. It performs first-pass data-quality checks and generates a cleaned output and quality report with one click. I tested it across different datasets to validate reusability.”

7. Moving to another laptop

- For personal use, transfer/import PERSONAL.XLSB and configure it in the new Excel installation.
- For sharing/demo, a macro-enabled .xlsm workbook is easier to package.

- For a more installable product, the same logic can later be packaged as an Excel Add-in (.xlam).
- Macro security may block VBA. Use approved/trusted settings rather than disabling security blindly.

8. Important limitation

This is a **generic first-pass data-quality cleaner**, not an AI or business-rule engine. It should not invent missing values or make domain-specific corrections without explicit rules. Always review the quality report and flagged records.

Universal_Data_Cleaner VBA Code

Paste this complete code into a module inside PERSONAL.XLSB. Keep a backup before editing.

Option Explicit

```
Sub Universal_Data_Cleaner()  
    Dim targetWB As Workbook, rawWS As Worksheet, cleanWS As Worksheet, reportWS As Worksheet  
    Dim lastRow As Long, lastCol As Long, r As Long, c As Long  
    Dim rawData As Variant, cleanData As Variant  
    Dim header As String, txt As String, flag As String  
    Dim missingCount As Long, duplicateCount As Long, flaggedRows As Long  
    Dim rowDict As Object, rowKey As String  
  
    Set targetWB = ActiveWorkbook  
    If targetWB Is Nothing Then MsgBox "Please open an Excel data file first.": Exit Sub  
    If UCase(targetWB.Name) = "PERSONAL.XLSB" Then MsgBox "Please select your data workbook first.": Exit Sub  
    Set rawWS = ActiveSheet  
    If Application.WorksheetFunction.CountA(rawWS.Cells) = 0 Then MsgBox "The active sheet is empty.": Exit Sub  
  
    lastRow = rawWS.Cells.Find(What:="*", SearchOrder:=xlByRows, SearchDirection:=xlPrevious).Row  
    lastCol = rawWS.Cells.Find(What:="*", SearchOrder:=xlByColumns, SearchDirection:=xlPrevious).Column  
    If lastRow < 2 Then MsgBox "At least one header row and one data row are required.": Exit Sub  
  
    Application.ScreenUpdating=False: Application.EnableEvents=False: Application.DisplayAlerts=False  
    rawData = rawWS.Range(rawWS.Cells(1,1), rawWS.Cells(lastRow, lastCol)).Value2  
    ReDim cleanData(1 To lastRow, 1 To lastCol+1)  
  
    For c=1 To lastCol  
        header=CleanText(CStr(rawData(1,c)))  
        If header="" Then header="Column_" & c  
        cleanData(1,c)=header  
    Next c  
    cleanData(1, lastCol+1)="Data_Quality_Flag"  
  
    For r=2 To lastRow  
        flag=""  
        For c=1 To lastCol  
            header=LCase(CStr(cleanData(1,c)))  
            If IsError(rawData(r,c)) Then  
                cleanData(r,c)="": AddFlag flag, "Excel Error: "&cleanData(1,c)  
            Else  
                txt=CleanText(CStr(rawData(r,c)))  
                If IsMissingValue(txt) Then  
                    cleanData(r,c)="": missingCount=missingCount+1  
                    AddFlag flag, "Missing: "&cleanData(1,c)  
                ElseIf InStr(header, "email")>0 Then  
                    txt=LCase(txt): cleanData(r,c)=txt  
                    If Not IsValidEmail(txt) Then AddFlag flag, "Check Email: "&cleanData(1,c)  
                ElseIf InStr(header, "phone")>0 Or InStr(header, "mobile")>0 Or InStr(header, "contact")>0 Then  
                    txt=DigitsOnly(txt): cleanData(r,c)=txt  
                    If Len(txt)<7 Or Len(txt)>15 Then AddFlag flag, "Check Phone: "&cleanData(1,c)  
                ElseIf InStr(header, "date")>0 Or header="dob" Or InStr(header, "birth")>0 Then  
                    If IsDate(rawData(r,c)) Then cleanData(r,c)=CDate(rawData(r,c)) Else cleanData(r,c)=txt: AddFlag flag, "Check Date: "&cleanData(1,c)  
                ElseIf IsFinancialColumn(header) Then  
                    If IsNumeric(CleanNumber(txt)) Then cleanData(r,c)=CDBl(CleanNumber(txt)) Else cleanData(r,c)=txt: AddFlag flag, "Check Number: "&cleanData(1,c)  
                Else  
                    If VarType(rawData(r,c))=vbString Then cleanData(r,c)=txt Else cleanData(r,c)=rawData(r,c)  
                End If  
            End If  
        Next c  
        If flag="" Then cleanData(r, lastCol+1)="Clean" Else cleanData(r, lastCol+1)=flag: flaggedRows=flaggedRows+1  
    Next r  
  
    Set rowDict=CreateObject("Scripting.Dictionary")  
    For r=2 To lastRow  
        rowKey=""  
        For c=1 To lastCol: rowKey=rowKey & "|" & CStr(cleanData(r,c)): Next c  
        If rowDict.Exists(rowKey) Then  
            duplicateCount=duplicateCount+1  
            If cleanData(r, lastCol+1)="Clean" Then  
                cleanData(r, lastCol+1)="Duplicate Record": flaggedRows=flaggedRows+1  
            Else  
                cleanData(r, lastCol+1)=cleanData(r, lastCol+1)&" | Duplicate Record"  
            End If  
        Else  
            rowDict.Add rowKey, True  
        End If  
    Next r  
  
    On Error Resume Next  
    targetWB.Worksheets("Clean_Data").Delete  
    targetWB.Worksheets("Data_Quality_Report").Delete  
    On Error GoTo ErrorHandler  
  
    Set cleanWS=targetWB.Worksheets.Add(After:=targetWB.Worksheets(targetWB.Worksheets.Count))  
    cleanWS.Name="Clean_Data"  
    cleanWS.Range(cleanWS.Cells(1,1), cleanWS.Cells(lastRow, lastCol+1)).Value=cleanData  
    With cleanWS: .Rows(1).Font.Bold=True: .Rows(1).AutoFilter: .Columns.AutoFit: End With  
  
    For c=1 To lastCol  
        header=LCase(CStr(cleanData(1,c)))  
        If InStr(header, "phone")>0 Or InStr(header, "mobile")>0 Or InStr(header, "contact")>0 Then  
            cleanWS.Columns(c).NumberFormat="@"  
        ElseIf InStr(header, "date")>0 Or header="dob" Or InStr(header, "birth")>0 Then  
            cleanWS.Columns(c).NumberFormat="dd-mm-yyyy"  
        ElseIf IsFinancialColumn(header) Then  
            cleanWS.Columns(c).NumberFormat="#.##0.00"  
        End If  
    Next c  
  
    Set reportWS=targetWB.Worksheets.Add(After:=targetWB.Worksheets(targetWB.Worksheets.Count))  
    reportWS.Name="Data_Quality_Report"  
    With reportWS  
        .Range("A1").Value="UNIVERSAL DATA QUALITY REPORT": .Range("A1").Font.Bold=True: .Range("A1").Font.Size=16  
        .Range("A3").Value="Metric": .Range("B3").Value="Result"  
        .Range("A4").Value="Source Workbook": .Range("B4").Value=targetWB.Name  
        .Range("A5").Value="Source Sheet": .Range("B5").Value=rawWS.Name  
        .Range("A6").Value="Rows Processed": .Range("B6").Value=lastRow-1  
        .Range("A7").Value="Columns Processed": .Range("B7").Value=lastCol  
        .Range("A8").Value="Missing Values": .Range("B8").Value=missingCount  
        .Range("A9").Value="Duplicate Records": .Range("B9").Value=duplicateCount  
        .Range("A10").Value="Rows Requiring Review": .Range("B10").Value=flaggedRows  
        .Range("A11").Value="Clean Output Rows": .Range("B11").Value=lastRow-1  
        .Range("A3:B3").Font.Bold=True: .Columns("A:B").AutoFit  
    End With  
    cleanWS.Activate
```

```

Application.DisplayAlerts=True: Application.ScreenUpdating=True: Application.EnableEvents=True
MsgBox "Universal Cleaning Completed!" & vbCrLf & vbCrLf & _
    "Rows Processed: "&Format(lastRow-1,"#,##0") & vbCrLf & _
    "Columns: "&lastCol & vbCrLf & _
    "Missing Values: "&Format(missingCount,"#,##0") & vbCrLf & _
    "Duplicates: "&Format(duplicateCount,"#,##0") & vbCrLf & _
    "Rows for Review: "&Format(flaggedRows,"#,##0"),vbInformation,"Universal Data Cleaner"
Exit Sub

ErrorHandler:
Application.DisplayAlerts=True: Application.ScreenUpdating=True: Application.EnableEvents=True
MsgBox "Error "&Err.Number&vbCrLf&vbCrLf&Err.Description,vbCritical,"Universal Data Cleaner"
End Sub

Function CleanText(ByVal txt As String) As String
    txt=Replace(txt,Chr(160)," "): txt=Replace(txt,vbTab," "): txt=Replace(txt,vbCr," "): txt=Replace(txt,vbLf," ")
    On Error Resume Next: txt=Application.WorksheetFunction.Trim(txt): On Error GoTo 0
    CleanText=txt
End Function

Function IsMissingValue(ByVal txt As String) As Boolean
    Select Case LCase(Trim(txt))
        Case "", "n/a", "na", "null", "none", "nil", "#n/a", "-": IsMissingValue=True
        Case Else: IsMissingValue=False
    End Select
End Function

Function DigitsOnly(ByVal txt As String) As String
    Dim i As Long, ch As String, result As String
    For i=1 To Len(txt): ch=Mid(txt,i,1): If ch>="0" And ch<="9" Then result=result&ch
    Next i
    DigitsOnly=result
End Function

Function CleanNumber(ByVal txt As String) As String
    txt=Replace(txt," ",""): txt=Replace(txt,"$",""): txt=Replace(txt,"■",""): txt=Replace(txt,"€",""): txt=Replace(txt,"£","")
    CleanNumber=Trim(txt)
End Function

Function IsFinancialColumn(ByVal header As String) As Boolean
    IsFinancialColumn=(InStr(header,"salary")>0 Or InStr(header,"revenue")>0 Or InStr(header,"amount")>0 Or InStr(header,"price")>0 Or InStr(header,"cost")>0 Or InStr(header,
End Function

Function IsValidEmail(ByVal email As String) As Boolean
    Dim atPos As Long, dotPos As Long
    atPos=InStr(1,email,"@"): dotPos=InStrRev(email,".")
    IsValidEmail=(atPos>1 And dotPos>atPos+1 And dotPos<Len(email))
End Function

Sub AddFlag(ByRef flags As String, ByVal newFlag As String)
    If flags="" Then flags=newFlag Else flags=flags&" | "&newFlag
End Sub

```